COMPILING THE ALMGREN–PITTS COMBINATORIAL LEMMA

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Abstract. The Almgren–Pitts combinatorial lemma is a recurring technical step in the min-max theory of minimal hypersurfaces. We extract it from the literature as a standalone metric-combinatorial algorithm and formalize it in Lean 4 as an importable library.

1. Introduction

A minimal hypersurface is a critical point of the area functional in codimension one. On a closed Riemannian manifold (M^{n+1}, g), existence of a nontrivial minimal hypersurface is nontrivial: direct minimization over a fixed homology class may produce only degenerate configurations, and the relevant critical points are typically of saddle type rather than minima. Birkhoff (1917) established the prototype of a *min-max* construction by producing closed geodesics on the sphere via a sweepout by loops; Almgren and Pitts, in the 1960s and 1970s, generalized the construction to higher-dimensional varifolds. The output is the following existence theorem, with regularity supplied by Schoen–Simon [SS81].

Theorem 1.1 (Almgren–Pitts min-max theorem [Pit81, SS81])**.** *Every closed Riemannian manifold (M^{n+1}, g) admits a nontrivial embedded minimal hypersurface $\Sigma \subset M$ whose singular set* $\operatorname{Sing}\Sigma$ *has Hausdorff dimension at most $n-7$.*

Theorem 1.1 was established in its systematic form by Pitts in the monograph [Pit81]; it has subsequently been re-proved and extended in various frameworks by Colding–De Lellis [CDL03], De Lellis–Tasnady [DLT13], Marques–Neves [MN14], and others.

Pitts's proof splits into two parts. The first part executes the min-max construction: starting from a sweepout family Λ and the min-max value

$$m_0(\Lambda) \;=\; \inf_{\Lambda} \max_{t} \mathcal{H}^n(\Gamma_t),$$

one produces an almost-minimizing sequence of varifolds. The second part is the regularity theory for the limiting varifold. Within the first part, one key technical step is a combinatorial assembly. In the formulation of De Lellis–Tasnady [DLT13, §3.2], it reads as follows.

Theorem 1.2 (De Lellis–Tasnady [DLT13, §3.2])**.** *Let $\{\partial\Omega_t\}_{t\in[0,1]}$ be a sweepout of (M^{n+1}, g) with $\mathcal{H}^n(\partial\Omega_t)$ continuous in t, let $m_0 = \sup_t \mathcal{H}^n(\partial\Omega_t)$, and let $N > 0$. Suppose that at each t with $\mathcal{H}^n(\partial\Omega_t) \geq m_0 - 1/N$ there is a nested pair $A_t \subset\subset B_t \subset\subset U'_t$ in M such that $\partial\Omega_t$ is almost minimizing in (A_t, B_t), and a replacement family $\{\partial\Omega_{t,\tau}\}_{\tau\in[0,1]}$ (given by* [DLT13, Lemma 3.1]*) with $\partial\Omega_{t,0} = \partial\Omega_t$ and $\mathcal{H}^n(\partial\Omega_{t,1}) \leq \mathcal{H}^n(\partial\Omega_t) - 1/(4N)$ inside U'_t. Then there exists a modified sweepout $\{\partial\Omega'_t\}$ with $\sup_t \mathcal{H}^n(\partial\Omega'_t) \leq m_0 - 1/(4N)$.*

The combinatorial content of Theorem 1.2 — a one-parameter assembly turning local near-critical savings into a global supremum improvement — recurs in Pitts's original proof of Theorem 1.1 [Pit81, proof of Thm. 4.10] and in the subsequent treatments [CDL03, §5], [DLR18, Prop. 4.8], [DL16, Prop. 5.8], and [MN14, §4], each time formulated in the geometric language of the paper at hand.

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The recurrence is structural. The statement of Theorem 1.2 is paper-agnostic: it depends on no geometric measure theory, no admissible class, no ambient regularity. Its content is a bookkeeping argument on continuous real-valued functions, free of the geometric material that varies between papers. Yet the statement has no canonical reference, so each author re-derives it in local notation.

This note takes a different approach. We extract Theorem 1.2 as an independent statement and formalize it in Lean 4 as the library `CombLemma`. Informally, the extracted form is the following.

Theorem 1.3 (Extracted combinatorial lemma; formalized as ✓ `exists_sup_reduction`)**.** *Let X be a pseudo-metric space equipped with an abstract paired-cover structure. Let $f : [0,1] \to \mathbb{R}$ be continuous with $m_0 = \sup f$ and $m_0 > 0$, and let $N > 0$. Assume that at every $t \in [0,1]$ with $f(t) \geq m_0 - 1/N$ there is a* local witness*: an open neighborhood $U_t \ni t$, a paired cover c_t in X, and a continuous replacement energy $E_t : [0,1] \to \mathbb{R}$ satisfying $f(s) - E_t(s) \geq 1/(4N)$ for all $s \in U_t$. Then there exists $f' : [0,1] \to \mathbb{R}$ with $\sup f' \leq m_0 - 1/(4N)$.*

Theorem 1.3 is the scalar skeleton of Theorem 1.2. The geometric content has been stripped: the boundary $\partial\Omega_t$ is replaced by a continuous function $f : [0,1] \to \mathbb{R}$; the Hausdorff-measure hypothesis becomes $m_0 = \sup f$; the nested pair (A_t, B_t, U'_t) collapses to a single open neighborhood U_t; the replacement family $\{\partial\Omega_{t,\tau}\}_{\tau\in[0,1]}$ collapses to a single continuous E_t; the almost-minimizing assumption on the varifold disappears. What remains is the combinatorial bookkeeping: a near-critical cover, uniform saving bounds, and the assembly of a globally improved f'. The abstract paired-cover class on X is a scaffolding parameter that would, in a GMT instantiation, carry the geometric data of the replacement; it is not load-bearing in the present proof (see §4). The formalization is machine-verified against `Mathlib`; it declares no custom axioms and contains no `sorry`. A subsequent min-max paper that establishes its own local-replacement lemma can then invoke the combinatorial step as a one-line import, with the verification following along.

§2 gives the library specification (interface, public API, internal lemmas); §3 sketches the proof; §4 records the structural details the formalization makes explicit.

2. Specification

We present the library specification following standard software documentation conventions: interface first (§2.1), public API (§2.2–§2.3), internal lemmas (§2.4).

2.1. Interface: types and structures.

Throughout, `K` denotes a topological space (the parameter space), `X` a pseudo-metric space (the ambient space whose covers witness replacements), and `f : K → ℝ` a continuous scalar energy.

Definition 2.1 (`PairableCover` class)**.** A `PairableCover X` instance declares, for each abstract `c : Cover`, a left region and a right region in X and a scalar diameter, subject to two axioms: the two regions are disjoint, and overlapping combined regions of diameter-comparable covers are ordered by inclusion.[1] Formal: ✓ `PairableCover`.

Definition 2.2 (`LocalWitness` structure)**.** A `LocalWitness K X f t` ε carries an open neighborhood $U \ni t$ in K, a cover `c : Cover`, a continuous replacement energy $E : K \to \mathbb{R}$, and the inequality $\forall s \in U,\ f(s) - E(s) \geq \varepsilon$. Formal: ✓ `LocalWitness`.

Definition 2.3 (Near-critical set)**.** For continuous $f : K \to \mathbb{R}$, scalar m_0, and $\delta \geq 0$, the δ-near-critical set is $\{t \in K : f(t) \geq m_0 - \delta\}$. When $\delta = 1/N$ we write NC. Formal: ✓ `nearCritical`.

Definition 2.4 (`FiniteCoverWithWitnesses` structure)**.** A `FiniteCoverWithWitnesses K f` m_0 δ ε carries a finite nonempty index type ι, pieces $\{p_l \subseteq K\}_{l\in\iota}$, replacement energies $\{E_l : K \to \mathbb{R}\}$, and savings $\{s_l \in \mathbb{R}_{>0}\}$, subject to four conditions: (I) $f - E_l \geq s_l$ on p_l; (II) each $t \in K$ lies in at most two pieces; (III) $s_l \geq \varepsilon$ for all l; (IV) $\{t \in K : f(t) \geq m_0 - \delta\} \subseteq \bigcup_l p_l$. Formal: ✓ `FiniteCoverWithWitnesses`.

[1] See Remark 4.1 on the scaffolding status of this class.

Definition 2.5 (`InitialCover` structure)**.** Specialized to $K = \mathbf{I}$. An `InitialCover X f m₀ N` carries a finite family of open intervals $I_i = (c_i - r_i, c_i + r_i) \cap \mathbf{I}$ for $i = 1, \ldots, n$, each paired with a witness center $w_i \in \mathrm{NC}$ and a local witness at w_i with saving $1/(4N)$, subject to (a) spacing: $c_i + r_i < c_{i+2} - r_{i+2}$ for all valid i; (b) containment: $I_i \subseteq U_{w_i}$; (c) coverage: $\mathrm{NC} \subseteq \bigcup_i I_i$.[2] Formal: ✓ `InitialCover`.

Definition 2.6 (`PartialRefinement` structure)**.** Given `ic : InitialCover` and $\ell \in \mathbb{N}$, a `PartialRefinement ic ℓ` carries pieces $J_1, \ldots, J_\ell \subseteq \mathbf{I}$ and an index assignment $\sigma : \{1, \ldots, \ell\} \to \{1, \ldots, n\}$ with (i) $J_k \subseteq I_{\sigma(k)}$ for each k; (ii) $I_{\sigma(k)} \subseteq \bigcup_{k'} J_{k'}$ for each k. Formal: ✓ `PartialRefinement`.

2.2. Public API: abstract core.

Theorem 2.7 (`exists_sup_reduction_of_cover`)**.**
Input. *A compact, nonempty topological space K; a continuous $f : K \to \mathbb{R}$; $m_0 = \sup f$; scalars $0 < \varepsilon \le \delta$; a `FiniteCoverWithWitnesses`$(K, f, m_0, \delta, \varepsilon)$ (Definition 2.4).*
Output. *There exists $f' : K \to \mathbb{R}$ with $\sup f' \le m_0 - \varepsilon$.*
Formal: ✓ `exists_sup_reduction_of_cover`.

2.3. Public API: one-parameter application.

Theorem 2.8 (`exists_sup_reduction`)**.**
Input. *A pseudo-metric space X with a `PairableCover X` instance (Definition 2.1); a continuous $f : \mathbf{I} \to \mathbb{R}$; $m_0 \in \mathbb{R}$ with $m_0 > 0$ and $m_0 = \sup f$; $N \in \mathbb{N}$ with $N > 0$; for every $t \in \mathrm{NC}$ a `LocalWitness` at t with saving $1/(4N)$ (Definition 2.2).*
Output. *There exists $f' : \mathbf{I} \to \mathbb{R}$ with $\sup f' \le m_0 - 1/(4N)$.*
Specializes Theorem 2.7 to $K = \mathbf{I}$, $\delta = 1/N$, $\varepsilon = 1/(4N)$. Formal: ✓ `exists_sup_reduction`.

2.4. Internal lemmas.

The declarations below are consumed by Theorem 2.8; proofs are sketched in §3.

Proposition 2.9 (Compactness of NC)**.** *If K is compact and f continuous,* NC *is compact. If additionally $m_0 = \sup f$ and $N > 0$,* NC *is nonempty. Formal: ✓ `isCompact_nearCritical`, ✓ `nearCritical_nonempty`.*

- ✓ `exists_initialCover` — existence of an `InitialCover` via a grid plus Lebesgue-number construction on the compact sublevel set NC.
- ✓ `exists_terminal_refinement` — existence of a terminal `PartialRefinement` with injective σ via bounded smallest-index induction with two-case dispatch.
- ✓ `InitialCover.not_three_overlap` — at most two among any three distinct `InitialCover` intervals overlap at a common point (parity rescue).
- ✓ `saving_bound_closure` — the open-neighborhood saving bound in `LocalWitness` extends to the closure via continuity of E.
- ✓ `Refinement.exists_refinement` — packages the 1D cover construction output as a `FiniteCoverWithWitnesses`$(\mathbf{I}, f, m_0, 1/N, 1/(4N))$, the input to Theorem 2.7.

3. Proof structure

We split the proof of Theorem 2.8 into three stages. Stage A builds an initial cover from the witness hypothesis. Stage B runs an induction producing a terminal partial refinement with injective σ. Stage C assembles the finite cover with witnesses by extending saving bounds to closures and deriving TwoFold. Theorem 2.7 closes.

[2] Conditions (a) and (b) transcribe DLT [DLT13, §3.2]'s spacing and subset conditions verbatim; the pair (c_i, w_i) replaces DLT's single t_i, see Finding 2 in §4.

3.1. Stage A: existence of the initial cover.

Proposition 3.1 (Existence of an initial cover)**.** *Under the hypotheses of Theorem 2.8, an initial cover (Definition 2.5) exists* ✓ `exists_initialCover`.

Construction. Compactness of NC (Proposition 2.9) plus the open cover $\{U_t\}_{t\in\mathrm{NC}}$ afforded by the witness hypothesis admit a Lebesgue number $\lambda > 0$: every λ-ball in NC fits inside some U_t. Choose $M \in \mathbb{N}$ with $M > 4/\lambda$, form the grid $c_k = k/M$ for $k = 0, \dots, M$, and select those c_k within $\lambda/4$ of NC. For each selected c_k pick a witness center $w_k \in \mathrm{NC}$ within $\lambda/4$ of c_k; the ball of radius λ around w_k fits inside some U_{w_k}, hence so does the interval $I_k := (c_k - 2/(3M), c_k + 2/(3M)) \cap \mathbf{I}$. Spacing condition (a) holds because consecutive selected grid indices differ by at least 2 in the selected finset, so their centers differ by at least $2/M > 4/(3M) = 2 \cdot (2/(3M))$. Coverage condition (c) holds because every $t \in \mathrm{NC}$ is within $1/(2M) < 2/(3M)$ of its rounded grid neighbor, which is selected. Condition (b) holds by the Lebesgue argument. □

3.2. Stage B: the refinement induction.

Starting from an initial cover, one builds pieces inductively.

Definition 3.2 (Base case)**.** step_zero(ic) is the length-1 partial refinement with $J_1 := I_1$ and $\sigma(1) := 1$ ✓ `step_zero`.

Definition 3.3 (Inductive step)**.** Given a partial refinement pr of length $\ell + 1$ and an index $i^* \in \{1, \dots, n\}$ with $I_{i^*} \not\subseteq \bigcup_k J_k$, define a length-$(\ell+2)$ extension by case split on whether $I_{i^*} \subseteq J_\ell \cup I_{\sigma(\ell)}$:

(Case 1) If yes, set $J_{\ell+1} := I_{i^*}$.
(Case 2) Otherwise, set $J_{\ell+1} := I_{i^*} \setminus I_{\sigma(\ell)}$.

In both cases set $\sigma(\ell + 1) := i^*$. Both invariants of Definition 2.6 are preserved; in Case 2 the preservation of (ii) uses the induction hypothesis on $I_{\sigma(\ell)} \subseteq \bigcup_k J_k$ ✓ `step_succ_at`.[3]

Proposition 3.4 (Termination)**.** *Starting from* step_zero(ic) *and iterating* step_succ_at*, one reaches in at most n steps a partial refinement of some length L with σ injective and $I_i \subseteq \bigcup_k J_k$ for every i* ✓ `exists_terminal_refinement`.

Argument. Let rem(pr) be the finite set of indices i with $I_i \not\subseteq \bigcup_k J_k$. One step of step_succ_at at $i^* \in \mathrm{rem}(\mathrm{pr})$ (chosen as the smallest such) produces pr′ with $i^* \notin \mathrm{rem}(\mathrm{pr}')$ and $\mathrm{rem}(\mathrm{pr}') \subseteq \mathrm{rem}(\mathrm{pr}) \setminus \{i^*\}$, hence strict decrease. Injectivity of σ follows from the case split in Definition 3.3: the new $\sigma(\ell + 1) = i^*$ is fresh because $i^* \in \mathrm{rem}$ while existing $\sigma(k)$ correspond to already-covered indices. □

3.3. Stage C: assembly.

The final stage extends the saving bound from open J_k to $\overline{J_k}$ and derives TwoFold from spacing (a).

Proposition 3.5 (TwoFold via parity)**.** *For a partial refinement* pr *of length L with σ injective, every $t \in \mathbf{I}$ lies in at most two of $\overline{J_1}, \dots, \overline{J_L}$* ✓ `terminal_twoFold`.

Argument. From spacing condition (a), iterated m times, one obtains that indices i, j with $|i - j| = 2(m + 1)$ (an even gap ≥ 2) have strict separation between $\overline{I_i}$ and $\overline{I_j}$, hence $\overline{I_i} \cap \overline{I_j} = \emptyset$ ✓ `InitialCover.chain_spacing`, ✓ `InitialCover.closure_disjoint_of_even_gap`.

Suppose three distinct indices $k_1, k_2, k_3 \in \{1, \dots, L\}$ admit $t \in \overline{J_{k_1}} \cap \overline{J_{k_2}} \cap \overline{J_{k_3}}$. Since $\overline{J_k} \subseteq \overline{I_{\sigma(k)}}$ and σ is injective, $\sigma(k_1), \sigma(k_2), \sigma(k_3)$ are three distinct indices in $\{1, \dots, n\}$ with a common point in the closures of $I_{\sigma(k_i)}$. Among any three distinct integers $a < b < c$ some pair has even gap ≥ 2: either $c - a$ is even (and necessarily ≥ 2), or $c - a$ is odd and ≥ 3, in which case $b - a$ and $c - b$ have opposite parities and one of them is the even gap. The pair with even gap has disjoint closures, contradicting the common point ✓ `InitialCover.not_three_overlap`. □

[3] A variant `step_succ` chooses i^* internally via Finset.min′; `step_succ_at` takes i^* as an explicit argument and returns the extension bundled with the "i^* is covered" property, for termination.

Proposition 3.6 (Closure extension of saving bound)**.** *Let* pr *be a partial refinement. For every k and every $t \in \overline{J_k}$, $f(t) - E_{\sigma(k)}(t) \geq 1/(4N)$, where $E_{\sigma(k)}$ is the replacement energy of the $\sigma(k)$-th local witness* ✓ `saving_bound_closure`.

Argument. On $J_k \subseteq U_{w_{\sigma(k)}}$ the witness gives $f - E_{\sigma(k)} \geq 1/(4N)$. The function $g := f - E_{\sigma(k)}$ is continuous (Definitions 2.2 requires E_t continuous; f is continuous by hypothesis). A closure point is a limit of a sequence in J_k, and g preserves $\geq$-inequalities under limits. □

Lemma 3.7 (Assembly)**.** *A partial refinement* pr *of length L with σ injective and $I_i \subseteq \bigcup_k J_k$ for every i yields a refinement of $(\mathbf{I}, X, f, m_0, N)$ in the sense of Definition 2.4, with $p_k := \overline{J_k}$, uniform saving $s_k := 1/(4N)$, and $E_k :=$ the replacement energy of the $\sigma(k)$-th witness* ✓ `Refinement.exists_refinement`.

Argument. Invariant (I) is Proposition 3.6. Invariant (II) is Proposition 3.5. Invariant (III) holds by construction ($s_k = 1/(4N)$). Coverage of NC chains $\mathrm{NC} \subseteq \bigcup_i I_i \subseteq \bigcup_k J_k \subseteq \bigcup_k \overline{J_k}$. □

Theorem 2.8 now follows: chain Propositions 3.1 and 3.4, invoke Lemma 3.7, and apply Theorem 2.7.

4. Discussion

4.1. **Eight structural points.** Formalizing the proof surfaces eight structural details that the prose exposition in [Pit81, CDL03, DLT13, DLR18, DL16, MN14] compresses. None is a correction; each is a place where the prose-level argument is locally underdetermined in a way the formal proof must resolve.

(1) The witness saving must commit to $\varepsilon = 1/(4N)$ exactly, not $\exists \varepsilon > 0$. The weaker hypothesis does not thread through the lower bound on the sum-of-savings in Theorem 2.7.
(2) The single t_i of DLT [DLT13, §3.2] must be decoupled into an interval center c_i and a witness center w_i (Definition 2.5). The existence proof with $c_i = w_i$ forced is intractable under the standard Lebesgue-number approach: the grid point c_k supplying the interval is not in general near-critical.
(3) The index assignment σ is not monotone (Definition 2.6, Proposition 3.4). Intuition from geometric pictures suggests it should be; the smallest-index-selection rule occasionally backtracks when an earlier index has been covered as a side effect of a later step.
(4) Spacing condition (a) yields disjointness of $\overline{I_i}$ and $\overline{I_j}$ only for even gaps $|i-j| = 2(m+1)$. A counterexample with non-monotone centers (permitted by the abstract structure) witnesses overlap at odd gap 3.
(5) TwoFold (Proposition 3.5) therefore requires an explicit parity argument: among three distinct indices some pair has even gap. DLT [DLT13, §3.2] asserts TwoFold in one sentence; the unpacking into chain-disjointness plus the parity rescue occupies a full proposition.
(6) The `PairableCover` class's `diameter_nesting` axiom is not invoked in the present proof. The scalar specialization to $\mathbf{I}$ uses explicit radii instead. See Remark 4.1.
(7) The replacement energy of a local witness must carry continuity explicitly (Definition 2.2). In GMT use this is automatic; under the abstraction, continuity is load-bearing in Proposition 3.6 and must be an explicit field of `LocalWitness`.
(8) The ambient class X and `PairableCover` are carried in signatures but not consumed by the proof chain; see Remark 4.1.

4.2. **Scaffolding.**

Remark 4.1 (`PairableCover` as scaffolding)**.** The `PairableCover` class is carried through `LocalWitness` and the intermediate structures but its axioms are not used in the proof of Theorem 2.8. Three outcomes are under consideration for v0.2: retain the class as an extension point (future abstract-K formulations may consume it); rework the proof so the class is load-bearing (following DLT's nested

pair structure $A \subset\subset B \subset\subset U'$ with annular shell $C = B \setminus A$); or remove it in favor of a scalar-only interface. The v0.1 release defers the decision.

4.3. **What is and is not in the repository.** `CombLemma` v0.1.0 contains the mathematics described above and nothing else. In particular:

- It does *not* construct local witnesses. The hypothesis of Theorem 2.8 is supplied by the consumer.
- It does *not* contain any GMT content. Varifolds, Caccioppoli sets, currents, measure-theoretic regularity theorems are all absent.
- It does *not* prove the full min-max theorem. The local-replacement lemma and the min-max closure remain the consumer's work.
- It does *not* address multi-parameter sweepouts. Generalization to K other than $\mathbf{I}$ is future work; see §4.4 below.

4.4. **Coverage across the min-max literature.** The formalization of `CombLemma` v0.1 specializes the sweepout parameter space to $K = [0, 1]$. Every min-max construction of minimal hypersurfaces that operates on a one-parameter sweepout consumes the same combinatorial assembly, and its scalar content is what Theorem 1.3 formalizes. The list of one-parameter constructions that `CombLemma` covers (as a citation replacing the paper's internal re-derivation of the combinatorial step) is summarized in Table 1.

Source	**Where the one-parameter sweepout appears**
Pitts [Pit81, proof of Thm. 4.10], $m = 1$ case	The $(m, \mathbf{M})$-homotopy class of the main existence theorem, specialized to $m = 1$.
Simon–Smith [Smi83]; Colding–De Lellis [CDL03, §5]	1-parameter sweepout of surfaces in 3-manifolds, with continuous refinement.
De Lellis–Tasnady [DLT13, §3.2]	"Standard notion of 1-parameter family of hypersurfaces," $\{\partial\Omega_t\}_{t\in[0,1]}$.
De Lellis–Ramic [DLR18, Prop. 4.8]	1-parameter sweepout with boundary.
Li–Zhou [LZ17]	Free-boundary 1-parameter sweepout $\phi : [0, 1] \to \mathcal{Z}_n(\Omega, \partial\Omega)$.
Guaraco [Gua18]	1-parameter Allen–Cahn min-max, $\Phi : [0, 1] \to W^{1,2}(M)$.
Ketover [Ket16]	Equivariant 1-parameter Simon–Smith min-max.
Chambers–Liokumovich [CL16]	1-parameter family of hypersurfaces for existence in complete, finite-volume manifolds.
Chambers–Liokumovich [CL20]	Monotonicity framework, $\{\partial\Omega_t\}_{t\in[0,1]}$.
Chodosh–Liokumovich–Spolaor [CLS22]	Non-excessive sweepout framework, explicitly "one-parameter min-max."
Zhou [Zho19]	Multiplicity-one conjecture for 1-parameter sweepouts.

TABLE 1. One-parameter min-max works whose combinatorial step is covered by `CombLemma` v0.1. In each, the paper's local-replacement lemma supplies the hypothesis of Theorem 1.3; the combinatorial assembly becomes a one-line invocation of `CombLemma.exists_sup_reduction`.

Multi-parameter extensions are *not* covered by v0.1. These include Pitts's general $(m, \mathbf{M})$ framework [Pit81, Thm. 4.10] with $m \geq 2$, the 5-parameter Willmore construction of Marques–Neves [MN14], the Weyl-law and density work of Liokumovich–Marques–Neves [LMN18] and Irie–Marques–Neves [IMN18], and De Lellis's multi-parameter exposition [DL16, Prop. 5.8]. A multi-parameter generalization is planned; see §4.5 for the specific interface changes it entails.

4.5. **Library roadmap.** Because `CombLemma`'s deliverable is a program and its public interface (the `LocalWitness` structure, the statement of `exists_sup_reduction`) is what downstream papers consume, a "future work" list for this paper is a versioning roadmap rather than a conjecture list. The planned trajectory is as follows.

- **v0.2 (generalize K).** Move the parameter space from `unitInterval` to an arbitrary compact metric space. The `LocalWitness` structure is unchanged. The `InitialCover` structure is redesigned: spacing condition (a), which currently relies on the linear order of $[0, 1]$ (indices i and $i + 2$), is replaced by a metric ball-covering condition on K (pairs of balls at distance $\geq \lambda$ in the metric). The parity-rescue argument of §3 Stage C becomes a disjointness argument on multiplicity in the metric cover. Downstream consumers who already operate on $[0, 1]$ keep their existing call sites unchanged.
- **v0.3 (multi-parameter $K = [0, 1]^m$).** Move from arbitrary compact K to a cubical-complex parametrization following Pitts [Pit81]. The `InitialCover` family of open intervals $\{I_i\}$ becomes a family of open cubes indexed by a cubical complex $I(m, n)$; the spacing condition becomes a face-relation bound; the parity argument becomes a cube-colouring argument counting cells by parity of coordinate-sum. `LocalWitness` and the main-theorem signature remain unchanged up to substituting K; existing downstream consumers continue to compile.
- `PairableCover` **decision.** The class is scaffolding in v0.1 (Finding 8, §4.1; Remark 4.1). v0.2 either (a) activates it by reworking Stage C (§3) to consume the diameter-nesting axiom in a DLT-style annular shell argument, or (b) removes it in favour of a scalar-only interface. Selection between (a) and (b) will be driven by whichever form a concrete downstream consumer establishes first.
- `Mathlib` **integration.** `CombLemma` is currently a standalone project with its own package name. If the material is suitable for upstreaming, a future version will adapt naming to Mathlib conventions (namespace layout, camel/snake case per declaration type, split granularity matching Mathlib's file-per-concept norm) and submit the library as a `Mathlib` pull request. The adaptation will not change the public API names; it will re-export them under Mathlib's namespace structure.

None of the above is a mathematical open problem. Each is a defined interface change with a concrete scope, a target version, and an identified downstream compatibility story.

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Appendix A. Repository structure

`CombLemma` is a Lean 4 `lake` project with a pinned `Mathlib` dependency (commit recorded in `lake-manifest.json`). The source layout is:

```
CombLemma.lean          top-level re-export
CombLemma/
  Witness.lean          PairableCover, LocalWitness
  Core.lean             FiniteCoverWithWitnesses, exists_sup_reduction_of_cover
  EnergyBound.lean      re-export stub (subsumed by Core as of v0.2)
  SupReduction.lean     exists_sup_reduction (one-parameter app)
  Refinement.lean       facade re-exporting six submodules
  Refinement/
    InitialCover.lean   nearCritical, InitialCover
    CoverConstruction.lean  exists_initialCover
    PartialRefinement.lean  PartialRefinement, step_zero
    Induction.lean      step_succ, step_succ_at, exists_terminal_refinement
    Disjointness.lean   chain_spacing, disjoint_of_even_gap,
                        closure_disjoint_of_even_gap, not_three_overlap
    Assembly.lean       terminal_twoFold, saving_bound_closure,
                        exists_refinement
test/Smoke.lean         trivial PairableCover on R
examples/MinimalUsage.lean  invocation pattern for consumers
```

Fourteen Lean 4 files. The `Core.lean` module is the abstract-core target; `SupReduction.lean` is a thin wrapper that composes the 1D cover construction in `Refinement/` with the core.

Appendix B. Build and axiom audit

The library builds with

```
lake exe cache get   # fetch Mathlib oleans (first run)
lake build           # build CombLemma
lake build test      # build smoke test
lake build examples  # build examples
```

No custom axioms are declared. The axiom audit

```
#print axioms CombLemma.exists_sup_reduction
```

reports dependence on three axioms: `propext`, `Classical.choice`, `Quot.sound`. These are Lean 4's foundational axioms; all three are used throughout `Mathlib`.

Appendix C. GMT ↔ abstract glossary

GMT / min-max term	`CombLemma` **correspondence**
Sweepout	Continuous $f : K \to \mathbb{R}$
Sweepout parameter space	K (specialized to $\mathbf{I}$)
Width of the sweepout	$m_0 = \sup f$
Near-critical slice	t with $f(t) \geq m_0 - 1/N$
Replacement family at t	Local witness at t (Definition 2.2)
Energy of the replacement	Replacement-energy function E_t
Quantitative saving	Threshold ε; here $\varepsilon = 1/(4N)$
Varifold / current / Caccioppoli set	Ambient X (abstracted away)
Paired-cover structure on X	`PairableCover` X
Improved sweepout	f' (conclusion of Theorem 2.8)
Min-max contradiction	Downstream consumer responsibility

Appendix D. Installation and invocation

To consume `CombLemma` in a downstream Lean 4 project, add to the project's `lakefile`:

```
require CombLemma from git
  "https://github.com/MathNetwork/comb_arg.git" @ "v0.1.0"
```

Run `lake exe cache get` and `lake build`. In consumer code:

```
import CombLemma

example
    {X : Type*} [PseudoMetricSpace X] [CombLemma.PairableCover X]
    {f : unitInterval -> R} (hf : Continuous f)
    {m0 : R} (hm_pos : 0 < m0) (hm : m0 = sSup (Set.range f))
    {N : N} (hN : 0 < N)
    (witness : forall t : unitInterval, f t >= m0 - 1/(N:R) ->
        Nonempty (CombLemma.LocalWitness unitInterval X f t
                    (1 / (4 * (N : R))))) :
    exists f' : unitInterval -> R,
        sSup (Set.range f') <= m0 - 1/(4 * (N : R)) :=
  CombLemma.exists_sup_reduction hf hm_pos hm hN witness
```

(ASCII substitutions for Unicode in the Lean 4 source: `R` $= \mathbb{R}$, `N` $= \mathbb{N}$, `->` $= \to$, `>=` $= \geq$, `<=` $= \leq$, `in` $= \in$.)

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